



Long-term forestation-deforestation dynamics in a socioeconomic perspective: The case of Italy, 1936–2018

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ABSTRACT

To explore long-term socio-ecological relationships at the local scale, this study investigates the evolution of forest cover (1936–2018) vis à vis population, settlements, and economic activities in Italy using geo-spatial information sources that combine environmental and socioeconomic indicators. Long-term forestation (or deforestation) dynamics were assessed in more than 8000 municipalities through a comprehensive analysis of the background socioeconomic context. A multivariate framework integrating descriptive statistics and maps, exploratory pairwise correlations, and a confirmative canonical correlation analysis, has discriminated five distinctive paths in forest cover change over time. Urbanization-driven deforestation in accessible and economically dynamic flat-hilly districts, and forestation after land abandonment in remote rural areas, have been the most frequent processes of landscape transformation in Italy. The empirical findings of this study reveal the complex interactions between forests and local communities, thus supporting the core assumption of the well-known Forest Transition Theory, a key research topic in forest assessment and policy. A peculiar contribution of this research lies in highlighting local specificities in forest dynamics, which are shown to depend strictly on the economic and demographic characteristics of local communities, as well as on the degree of land protection. Triggering a path of sustainable development in rural areas requires policies that combine factors potentially beneficial for the environment, such as the expansion of forests, in order to create a more balanced set of socio-ecological conditions. This will help reducing depopulation and fostering local economies through low-impact socio-demographic processes that are compatible with the environmental quality typical of 'forest dominated' landscapes.

1. Introduction

Human-driven landscape transformations are often recognized as the main cause of environmental degradation, and usually lead to habitat fragmentation, biodiversity loss, and land degradation, among others (e.g. Baptista and Rudel, 2006; Blondel et al., 2010). However, empirical knowledge relating environmental change to socioeconomic dynamics is still required when designing policies for sustainable land management and rural development (Uusivuori et al., 2002). As a matter of fact, it is highly recommended that policy-making for sustainable development considers the multiple interactions between natural resources, human well-being, and economic prosperity in local systems (Zhang et al., 2006). A 'holistic' understanding of the complex relationship between

socioeconomic processes and ecosystem evolution provides essential guidance in land-use planning and habitat conservation initiatives, as well as in social and demographic policies aimed at territorial cohesion (Vedeld et al., 2007). Quantifying landscape characteristics and estimating environmental quality are therefore essential tasks when planning natural resources and outlining biodiversity conservation actions (Sirami et al., 2010). With this perspective in mind, conservation of high-quality natural environments (e.g. forests, shrubland, pastures, wetlands) require novel approaches that account for their joint socioeconomic and landscape value (Marchetti et al., 2014).

To investigate changes in natural resources – such as forest cover – in relation with socioeconomic contexts, such approaches should benefit from multiple insights from ecology, economics and geography, thus

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incorporating the best assessment practices from each discipline (Cimini et al., 2013). A broad spectrum of human processes has been associated with forestation and deforestation. Rural poverty, land abandonment and soil erosion basically shaped forestation patterns in both advanced economies and emerging countries (Corona et al., 2023). Industrialization, agricultural intensification, and global trade fuelled (directly or indirectly) deforestation (Raghavan and Shrimali, 2015). The impact of socioeconomic processes on forest ecosystems has also been studied in peri-urban contexts, where land-use changes may significantly influence ecosystem features such as tree cover (Dobbs et al., 2011).

Over the last century, drastic land-use transformations have been observed across Europe, determining a negative impact on natural ecosystems (European Environment Agency, 2006a). Various environmental and socioeconomic factors have been implicated in such processes (Lazarus, 2014), sometimes acting as the most significant drivers of change (Santos et al., 2014). Earlier studies have shown how land-use change, rather than deforestation per se, results in fragmentation, simplification and degradation of natural landscapes (Zhang et al., 2006). Furthermore, especially with accelerated population growth (e.g. between the two World Wars of the last century), wealthier countries generally demonstrated a shift from net-loss to net-expansion of forest area (Klooster, 2003; Bray and Klepeis, 2005; Rudel et al., 2005). To clarify the latent relationship between socio-demographic shifts and forest transition in advanced economies (and, particularly, in Europe), further research is required to diachronically analyze and compare multiple socioeconomic dynamics with respect to environmental change, since the empirical results available so far are still conflicting (Antrop, 2005).

The long-term effects of local socioeconomic structures on forest cover, and more recent changes in forest area, have been widely investigated and discussed in earlier studies, proposing original interpretations to complex socio-environmental systems (e.g. Barbati et al., 2013). One research approach to these issues is the Forest Transition Theory (FTT), a key framework for identifying and explaining long-term changes in forest cover and the underlying socioeconomic drivers (e.g. Mather and Needle, 1998; Mather et al., 1998, 1999). FTT argues that place-specific factors and the background context play a role in determining forest cover dynamics, especially over sufficiently long time scales (DeFries and Pandey, 2010). For instance, rapid urbanization and industrialization can fuel deforestation (Zambon et al., 2017), while policies promoting sustainable agriculture and the expansion of mixed (crop/forest) landscape mosaics can lead to progressive forest recovery (Perz, 2007).

By providing a starting point for the investigation of (apparent and latent) relationships between forest cover and socioeconomic indicators (Mather, 1992), FTT assumes that forest cover follows a U-shaped pattern over time (Mather, 2004). It initially declines as regional development and population growth lead to deforestation and then recovers with economic growth, wealth accumulation, and a greater awareness of local communities regarding environmental issues. Depending on location-specific socio-environmental conditions, FTT describes a long-term process with variable trends in forest cover, from net-loss to net-gain of forest area, and vice versa (Yackulic et al., 2011; Redo et al., 2012; Gutiérrez Rodríguez and Ruiz Pérez, 2013). Changes from net deforestation to net reforestation definitely reflect regional economic factors, population growth, urbanization, agricultural intensification, the spatially varying level of human capital, education and job skills, quality of governance, and environmental awareness, among others (de Jong, 2009; Barbier et al., 2010; Pagnutti et al., 2013). Empirical findings in line with FTT suggest the importance of local contexts in shaping the rate of woodland expansion (or decline) over sufficiently long time intervals (Colantoni et al., 2017), possibly delineating specific territorial profiles as characteristic of landscapes with net gain (or loss) in forest cover (Wilson et al., 2017).

Despite its simplified assumptions, FTT may inform policies introducing and promoting sustainable management of forest land (Mather

et al., 1998), and also advances our in-depth understanding of land-use change drivers (Ferrara et al., 2014; Otero et al., 2015; Salvati et al., 2015). Because they involve multiple dimensions that should be seen as synergistic factors of change, the close linkage between socio-demographic dynamics and forest transition, which have been widely studied in some developing regions of the world (e.g. Mather, 2004), are also relevant in advanced economies. Although some criticisms emerged against the stylized facts of the FTT, this theory provides an important framework when examining the relationship between forest cover and socioeconomic indicators. For this reason, FTT was adopted as the basic framework of our empirical analysis of long-term forest dynamics in Italy and the relationship with the local context.

Studying the complex dynamics that occur at the local scale can provide valuable insights into the specific factors and processes that drive forest transitions (He et al., 2014). More specifically, empirical research with a local focus helps identifying competing interests and conflicts between different stakeholders in land-use decision-making (Corona et al., 2007). This could facilitate the development of targeted interventions and policies that address the specific challenges and opportunities faced by different local communities (Corbelle-Rico et al., 2012). Furthermore, local-scale dynamics considers the unique socio-economic context of each region (Long et al., 2020), including factors such as population growth (or decline), the rise of the middle class in disadvantaged territories, the greater availability and use of digital technologies and the influence of cultural norms and traditional practices (Theobald, 2004).

A refined understanding of such dynamics is critical to understand system's complexity and address the intricacies of forest cover change (Crespo Cuaresma et al., 2017), enabling a more nuanced and context-specific analysis that informs effective strategies to promote sustainable forest management and conservation (Long and Qu, 2018). Despite being perceived as an important dimension of environmental dynamics (Trumbore et al., 2015), the geographical distribution of forest cover and the socioeconomic characteristics of local communities have received little attention (Zhang et al., 2006). Due to joint demographic, cultural, political, and economic processes, local communities reflect the long-term interaction between humans and natural resources (Salvati and Carlucci, 2011), promoting feedbacks, influencing thresholds, and finding synergistic relationships in complex socio-ecological systems (Kelly et al., 2015).

Furthermore, although some criticisms have arisen about the use of administrative boundaries in the quantification of biophysical variables (Li et al., 2021), earlier studies (e.g. Fares et al., 2017; Ferrara et al., 2017; Tomao et al., 2017) have proposed and tested municipalities as an elementary spatial domain that allows for a comprehensive profile of local communities with respect to the quality of a given territory. As fundamental units with distinctive features that satisfy data integration requirements across environmental, socio-demographic and economic fields, municipalities play a direct role in land management plans (Corona et al., 2008). This dimension allows inferring the impact of environmental degradation and land quality on production and socio-spatial structures (Corona et al., 2012), as well as the identification of pertinent geographical gradients that are assumed to play a role in the spatial organization of a given region (Colantoni et al., 2015). This approach is a compromise that allows the analysis of regions with different characteristics, but with the same national-level policy and planning system (Salvati et al., 2018), as well as the analysis of biophysical systems that evolve at significantly faster (or slower) rate than the socioeconomic system (Salvati and Carlucci, 2011).

Based on these premises, our study analyses the long-term evolution (1936–2018) of forest cover in Italy at municipal level, using a multi-disciplinary approach that combines environmental and socioeconomic information (Sabbì and Salvati, 2014). An in-depth examination of forest dynamics and management allows to identify, in any local context, relevant shifts in the use of forest resources from profit-making purposes to less profitable environmental or landscape purposes, e.g. aesthetic,

recreational, or conservative (Hajjar et al., 2014). By uncovering the complex interactions between forests and human society (Purves and Pacala, 2008), the knowledge framework resulting from this study (e.g. Corona et al., 2008; Crespo Cuaresma et al., 2017; Kauppi et al., 2018) can devise strategies to promote sustainable resource management and resilient communities.

2. Methodology

2.1. Study area

Italy, located between 36° and 47°30' North and 5°30' and 18°30' East, extends a total area of 301,330 km² (23 % lowlands, 42 % uplands, 35 % mountains), with the presence of two major mountain systems: the Alps in Northern Italy and the Apennines in Central-Southern Italy. The dominant climate regime is Mediterranean temperate with dry summers and wet winters; however, continental traits can be found in the local climate of some Northern Italian districts, especially across the Alpine arc (Incerti et al., 2007). Based on the European Commission's Habitat Directive no. 43/1992, one-third (32 %) of Italian forests are located in the alpine bio-geographical zone, 16 % in the continental region, and 52 % in the dry Mediterranean region. Seven out of the 14 European Forest Types (Barbati et al., 2014) are represented in Italy. Oaks (*Quercus* spp.) and Beech (*Fagus sylvatica*) are the dominant species in deciduous forests. Norway spruce (*Picea abies*), Mountain pines (*Pinus sylvestris* and *Pinus nigra*), and Mediterranean pines (*Picea halepensis*, *Pinus pinaster*, and *Pinus pinea*) are the most abundant species in coniferous forests (Vacchiano et al., 2012). Coppice is a silvicultural practice adopted in several broad-leaved forests (Ciancio et al., 2006). Making use of a relatively large number of indicators from multiple data sources (e.g. Biasi et al., 2015), our study involved 8101 elementary spatial units corresponding to the entire set of Italian municipalities as currently delineated by the Italian National Statistics Institute (ISTAT).

2.2. Data sources and elementary indicators

2.2.1. Forest indicators

The study refers to digital maps that assess the state (and basic characteristics) of forest cover in Italy at the beginning (1936) and the end (2018) of the study period. The first data source is the Italian Kingdom Forest Map made available thanks to a recovery and restoration project of the original map published in 1936 (Ferretti et al., 2018). The latter data source is the Corine Land Cover map dated 2018 (European Environment Agency, 2019). These two maps were created with the same declared resolution (1:100,000) and their comparability has been demonstrated, among others, by Camarretta et al. (2018). The topological overlay of the forest maps described above with the shapefile representing the boundaries of the Italian municipalities allows the construction of 16 indicators based on the 'zonal statistics' tool (Salvati et al., 2015) provided with the ArcGIS 9.3 software (ESRI Inc., Redwoods, USA). Eight indicators refer to the actual forest cover (2018 status) and the remaining 8 indicators quantify the dynamics of forest gain and loss between 1936 and 2018 (Table 1 and SM.Table 1). Among the indicators relating to the state of forest cover in 2018 is G1, which quantifies the percentage of forest cover with respect to the related municipal area. Using the same ratio, G2 describes the quantity of degraded forests, while G3, G4, G5, G6, and G7 refer to different forest types providing the relative percentage for each forest type considered (respectively, mixed stands, chestnuts, conifers, beeches, oaks) established with the aim at maintaining full comparability between the two data sources (1936 and 2018 maps). Finally, G8 estimates forest diversity by considering the Pielou J evenness index applied to the six forest classes mentioned above (from G2 to G7), indicating the distribution pattern of these species/groups. The indicators assessing changes in forest cover over time include H1 (percent change of total forest cover in the municipal area between 1936 and 2018), H2-H7 (percent change,

Table 1

Description of forest indicators and the socioeconomic dimensions considered here.

Acronym	Name	Dimension	Source
<i>Forest</i>			
G1	Forest covered area (%)	Forest cover	Corine Land Cover 2018 (EEA, 2019)
G2	Degraded forests (%)	Forest cover	
G3	Other forest types (%)	Forest cover	
G4	Chestnut (%)	Forest cover	
G5	Conifer (%)	Forest cover	
G6	Beech (%)	Forest cover	
G7	Oak (%)	Forest cover	
G8	Forest diversity index (J Pielou)	Forest cover	
H1	Forest covered area change (%)	Forest cover change	CLC 2018 (EEA, 2019); (Ferretti et al., 2018)
H2	Degraded forest change (%)	Forest cover change	
H3	Other forest types change (%)	Forest cover change	
H4	Chestnut cover change (%)	Forest cover change	
H5	Conifer cover change (%)	Forest cover change	
H6	Beech cover change (%)	Forest cover change	
H7	Oak cover change (%)	Forest cover change	
H8	Forest diversity index (J Pielou) variation	Forest cover change	
<i>Population and territory</i>			
I1	Total municipal footprint (km ²)		Censuses of population, agriculture, industry
I2	Non-occupied houses (%)	Human settlements	Census of population
I3	Average house size per inhabitant (m ²)	Human settlements	
I4	Compact urban settlements (% in total area)	Human settlements	European Environment Agency
I5	Population residing in urban centers and nuclei (%)	Human settlements	Census of population
I6	Dwellings built-up later than 1991 (% in total dwellings)	Human settlements	Census of buildings
I7	Property dwellings (% in total dwellings)	Human settlements	
I8	Average surface of dwellings (m ²)	Human settlements	
I9	Head municipality of a Travel-to-work district	Human settlements	Territorial statistics
I10	Self-commuting (% people working/residing in the same munic.)	Human settlements	Census of population
B1	Average municipal surface area (km ²)	Topography and land	Territorial statistics
B2	Perimeter to Area ratio	Topography and land	Istat (2006)
B3	Latitude gradient (0: north-center, 1: south Italy)	Topography and land	
B4	Proximity to the sea coast (0: inland; 1: coastal)	Topography and land	
B5	Land Quality Index	Topography and land	Europ. Environ. Agency
P1	Number of components per family	Demographic structure	Census of population
P2	Population > 80 years / births		
P3	Population > 75 years (%)	Demographic structure	

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Table 1 (continued)

Acronym	Name	Dimension	Source
P4	Elderly index	Demographic structure	Population register
P5	Dependency ratio	Demographic structure	
P6	Resident foreign people per 100 inhabitants	Demographic structure	
P7	Masculinity ratio	Demographic structure	
P8	Migration component of demographic balance (%)	Demographic structure	Census of population
P9	Families with children (% on total families)	Demographic structure	
P10	Civil weddings (% on total weddings)	Demographic structure	Population register
D1	Annual population growth rate (%), 1921–1931	Long-term demographic dynamics	Census of population
D2	Annual population growth rate (%), 1931–1936	Long-term demographic dynamics	
D3	Annual population growth rate (%), 1936–1951	Long-term demographic dynamics	
D4	Annual population growth rate (%), 1951–1961	Long-term demographic dynamics	
D5	Annual population growth rate (%), 1961–1971	Long-term demographic dynamics	
D6	Annual population growth rate (%), 1971–1981	Long-term demographic dynamics	
D7	Annual population growth rate (%), 1981–1991	Long-term demographic dynamics	
D8	Annual population growth rate (%), 1991–2001	Long-term demographic dynamics	
D9	Annual population growth rate (%), 2001–2011	Long-term demographic dynamics	
D10	Annual population growth rate (%), 2011–2020	Long-term demographic dynamics	
<i>Labour market and human capital</i>			
L1	Participation rate	Job market	Census of population
L2	Activity rate	Job market	
L3	Unemployment rate	Job market	
L4	Unemployment rate of young people (< 35 years)	Job market	
L5	Female participation rate	Job market	Census of Industry and Services
L6	Female activity rate	Job market	
L7	Female unemployment rate	Job market	
L8	Unemployment rate of young women (< 35 years)	Job market	
L9	Employees on total workers (%)	Job market	
L10	Women workers on total workers (%)	Job market	
L11	Consultants on total workers (%)	Job market	
L12	Temporary workers on total workers (%)	Job market	
L13	Apprentices on total workers (%)	Job market	
L14	Temporary workers on consultants (%)	Job market	
F1	Population with tertiary education (%)	Education	Census of population

Table 1 (continued)

Acronym	Name	Dimension	Source
F2	Population graduated in high-school (%)	Education	Census of Industry and Services
F3	Population with secondary education (%)	Education	
F4	Population with primary education (%)	Education	
F5	Literate population without formal education degree (%)	Education	
F6	Illiterate population (%)	Education	
<i>Economy</i>			
S1	Average number of workers per industrial local unit	Economic structure	Census of Industry and Services
S2	Density of workers per municipality surface area (km ²)	Economic structure	
S3	Workers in the agricultural and forestry sectors (%)	Economic structure	
S4	Workers in fishing and complementary activities (%)	Economic structure	
S5	Workers in the mining sector (%)	Economic structure	
S6	Workers in manufacturing (%)	Economic structure	
S7	Workers in energy production and distribution industries (%)	Economic structure	
S8	Workers in constructions (%)	Economic structure	
S9	Workers in the commerce sector (%)	Economic structure	
S10	Workers in hotels and restaurants (%)	Economic structure	
S11	Workers in transportation and logistics (%)	Economic structure	
S12	Workers in financial, insurance and banking services (%)	Economic structure	
S13	Workers in informatics, renting and real estate services (%)	Economic structure	
S14	Workers in the public sector (%)	Economic structure	
S15	Workers in education services (%)	Economic structure	
S16	Workers in the health sector (%)	Economic structure	
T1	Number of beds in hotels and campings / resident population	Tourism	Census of Industry and Services
T2	Average number of beds per hotel	Tourism	
T3	Hotel occupancy level (five-years average)	Tourism	Istat (2006)
T4	Camping occupancy level (five-years average)	Tourism	
T5	Agri-tourism occupancy level (five-years average)	Tourism	
T6	Number of beds in agri-tourism accomodation / beds in hotel	Tourism	
T7	Resident population / total number of stores	Tourism	
<i>Quality of life</i>			
Q1	Subscriptions on state radio-television channels (%)	Wealth	Banca d'Italia and Istituto Tagliacarne

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Table 1 (continued)

Acronym	Name	Dimension	Source
Q2	Number of cars / inhabitants	Wealth	Istituto Tagliacarne
Q3	Number of deposits / banks	Wealth	
Q4	Number of deposits / inhabitants	Wealth	
Q5	Value of bank deposits / banks (euros)	Wealth	
Q6	Average value of bank deposits (euros)	Wealth	
Q7	Value of bank deposits / inhabitants (euros)	Wealth	
Q8	Per capita income tax amount (euros)	Income	
Q9	Per capita real estate tax amount (euros)	Income	
Q10	Per capita municipal solid waste tax amount (euros)	Income	
Q11	Per capita disposable income (euros)	Income	
Q12	Per capita consumption (euros)	Income	
Q13	Total value added per municipality (euros)	Income	CENSIS
C1	Crime intensity index	Crime	Istat (2006)
C2	Crime severity index	Crime	
C3	Number of crimes per 1000 inhabitants	Crime	
C4	Work accidents per 100 inhabitants	Crime	
C5	Car accidents/total vehicles	Crime	
Agriculture			
SR-A1	Rented agricultural surface area / total agricultural surface area (%)	Land tenure	Census of agriculture
SR-A2	Agricultural land owned by the state (%)	Land tenure	
SR-A3	Average farm size (hectares)	Land tenure	
SR-A4	Total agricultural land / Total municipal surface area (%)	Land tenure	
SR-A5	Agricultural utilized area / Total agricultural land (%)	Land tenure	
SR-A6	Pielou's evenness on farm size distribution	Land tenure	
SR-Q1	Agricultural utilized area under organic farming (%)	Innovation and quality and innovation	
SR-Q2	Area cultivated with DOC designation of origin grapevines (%)	Innovation and quality and innovation	
SR-Q3	Area cultivated with DOCG designation of origin grapevines (%)	Innovation and quality and innovation	
SR-Q4	Livestock organic farms / Total farms (%)	Innovation and quality and innovation	
SR-Q5	Agricultural utilized area under good agronomic practices (%)	Innovation and quality and innovation	
SR-Q6	Agricultural utilized area under sustainability certification (%)	Innovation and quality and innovation	
SR-Q7	Density of cattle (%on total agricultural utilized area)	Innovation and quality and innovation	
SR-Q8	Agricultural utilized area applying sustainable irrigation (%)	Innovation and quality and innovation	

Table 1 (continued)

Acronym	Name	Dimension	Source
SR-Q9	Index of economic marginalization of farms	Innovation and quality and innovation	
SR-Q10	Density of agro-tourism activities (farms/ha)	Innovation and quality and innovation	
SR-L1	Employees in the primary sector (%)	Human capital	
SR-L2	Farmholders >55 years (%)	Human capital	
SR-L3	Farmholders on total workers in the primary sectors (%)	Human capital	
SR-L4	Farmholders with technical (agronomy) education (%)	Human capital	
SR-L5	Farmholder's activity diversification index	Human capital	
SR-L6	Family farms (%)	Human capital	
SR-P1	Agricultural utilized area under environmental protection (%)	Agricultural landscape	
SR-P2	Arable land / Agricultural utilized area (%)	Agricultural landscape	
SR-P3	Perennial crop / Agricultural utilized area (%)	Agricultural landscape	
SR-P4	Pastures and meadows / Total agricultural land (%)	Agricultural landscape	
SR-P5	Farm size diversity (Shannon index)	Agricultural landscape	
SR-P6	Farm surface area covered by woodland (% on total farm area)	Agricultural landscape	
SR-P7	Change in agricultural utilized area (1990–2000, %)	Agricultural landscape	
SR-P8	Agricultural landscape diversity (Shannon index)	Agricultural landscape	
SR-M1	Number of agricultural machines per farm	Crop intensity	
SR-M2	Irrigated land / Total agricultural utilized area (%)	Crop intensity	
SR-M3	Agricultural utilized area per worker in agriculture (ha)	Crop intensity	
SR-M4	Crop intensity index	Crop intensity	
<i>Environment</i>			
W1	Per capita distributed water	Water use/management	Census of water resources
W2	Water dispersion index	Water use/management	
W3	Consumed water/inhabitants	Water use/management	
W4	Proportion of water distributed to civil uses	Water use/management	
W5	Number of reservoirs (per 100 inhabitants)	Water use/management	
W6	Reservoir capacity (per 100 inhabitants)	Water use/management	
E1	Protected land (% in total municipal area)	Ecological quality	Italian Ministry of the Environment Salvati et al. (2011)
E2	Potential soil contamination footprint (equiv. inhab/km ²)	Ecological quality	
E3	Soil erosion risk index	Ecological quality	European Soil Bureau (JRC)
E4	Topsoil organic carbon content (%)	Ecological quality	
E5	Agricultural pressure to soil compaction index	Ecological quality	

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Table 1 (continued)

Acronym	Name	Dimension	Source
E6	Environmental pressure from grazing index	Ecological quality	European Soil Bureau (JRC)
E7	Landslide risk index	Ecological quality	European Soil Bureau (JRC)
E8	Average soil ph	Ecological quality	Salvati et al. (2011)
E9	Natural susceptibility to soil compaction index	Ecological quality	
E10	Soil organic carbon stock of agricultural soils index	Ecological quality	
E11	Soil salinization vulnerability index	Ecological quality	Salvati et al. (2011)
E12	Soil salinization risk index	Ecological quality	European Soil Bureau (JRC)
A1	Costs for (re)forestation of fire-affected woodland (euros per km ²)	Environmental management	Italian Forest Service ISTAT
A2	Number of fire events in total municipal area	Environmental management	Environmental management
A3	Environmental Sensitive Area Index (ESAI), 1960	Environmental management	
A4	Environmental Sensitive Area Index (ESAI), 1960–1990	Environmental management	
A5	Environmental Sensitive Area Index (ESAI), 1990–2000	Environmental management	Environmental management
A6	Environmental Sensitive Area Index (ESAI), 1990–2010	Environmental management	

respectively, of degraded forests, other forest types, chestnuts, conifers, beeches, oaks) and H8 (percent change in J Pielou evenness index).

2.2.2. Socioeconomic and territorial indicators

The individual variables considered in the present study were made available from official sources (Istat, 2006) and are fully described in SM.Table 1. For each Italian municipality, 154 indicators were computed from the variables collected and classified into six themes (population and territory: 35 indicators; labor market and human capital: 20; economy: 23; quality of life: 18; agriculture: 34; environment: 24), further organized into 19 dimensions (human settlement, topography and land, demographic structure, long-term population dynamics, job market, education, economic structure, tourism, wealth, income, crime, land tenure, innovation and quality, human capital in agriculture, agricultural landscape, crop intensity, water use and management, ecological quality, and environmental management). According with Salvati and Carlucci (2011), Sabbi and Salvati (2014), and Sallustio et al. (2018), the considered indicators, research dimensions and themes, were chosen to provide a comprehensive socioeconomic profile of local communities and a detailed analysis of the environmental contexts in Italy in relation to the state of forest cover and its change over time.

2.3. Data analysis

In the first phase of this work, we calculated descriptive statistics through a classification of municipalities based on the presence of forest cover in 1936 and on the changes that occurred between 1936 and 2018. Municipalities were therefore divided into five categories: (i) without forests in both 1936 and 2018; (ii) without forests in 1936 and with forests in 2018 (complete forestation); (iii) with forests in both 1936 and 2018, and with forest increment between 1936 and 2018 (partial forestation); (iv) with forests in both 1936 and 2018, and with forest decrease between 1936 and 2018 (partial deforestation); (v) with forests in 1936 and without forests in 2018 (total deforestation). For each of these categories, the average figures of selected territorial, demographic, and socioeconomic variables were evaluated. Simple correlations were run pair-wise, without considering the distinction of variables into different themes and dimensions. We adopted two-stage

non-parametric correlations (Spearman rank coefficients) identifying both linear and non-linear relationships typical of environmental-economic issues. The first step involved a dataset consisting of forest cover indicators (2018) and the related background indicators. A second dataset was used to delineate the latent relationship between forest change indicators (1936–2018) and the same background indicators that reflect dominant conditions during the study period.

The final phase of the study involved a Canonical Correlation Analysis (CCA), a multivariate technique aimed at finding multiple sets of reciprocally correlated variables, controlling for redundancy and thus extracting the most parsimonious representation of latent relationships between two complex datasets. The analysis was run twice, separately considering the sets of (i) 8 forest cover indicators, and (ii) 8 forest cover change indicators. In both cases, the analysis was run separately on the standardized matrix (i.e. leading to a parametric CCA based on Pearson correlations) and on rankings (i.e. leading to a non-parametric CCA based on Spearman correlations). CCA provides reliable results when multi-collinearity among predictors is relevant (Salvati and Zitti, 2009). CCA maximizes covariance between two sets of covariates (i.e. forest indicators and socioeconomic/territorial indicators) on the same observation conditions (i.e. municipal units). Explained covariance is calculated as the amount of covariance explained by each extracted axis, as a percentage of the total covariance. Since this descriptive analysis was not founded on hypothesis testing, the selection of significant roots was based on a-priori thresholds in accordance with previous studies (e.g. Ferrara et al., 2017). The canonical roots (also called canonical eigenvalues) are key values that represent the intensity of the relationship between two sets of multivariate variables. The CCA results allow an explicit evaluation of the position of each indicator (forest and context) and case (municipalities). Indicators' coefficients and municipalities were examined together with reference to the extracted canonical roots. To simplify the description of the results, only significant coefficients were considered (Salvati and Carlucci, 2016). A similar coefficient (both intensity and sign) on the same Root 1 indicates a convergent relationship between forest and context indicators. Similar coefficient intensity with an inverse sign instead represents an inverse relationship between forest conditions and territorial contexts.

3. Results

The classification of Italian municipalities according to forest cover (2018) and changes (1936–2018) suggests that municipalities experiencing 'partial forestation' dynamics are dominant in the Italian landscape (Table 2) and have the largest administered area, on average. They constitute more than half of the country area and more than 80 % of the national forest area (Fig. 1). Despite having the lowest population density, on average, these municipalities host almost 40 % of total population and have a lower than average per-capita income. Within this class, forest area has increased by nearly 170 %, being uniformly distributed throughout the Italian territory (480 m above sea level, on average). On the contrary, the 'total forestation' paths characterize a marginal part of municipalities which represent 5 % of the total forest area and 3 % of the country area, being concentrated in Southern Italy and along the sea coastline, and presenting the lowest per-capita income and the highest population density, on average (See Tables 3–5).

Municipalities without forests in both years 1936 and 2018 are mainly located in the Po plain and along the Adriatic Sea coast. They account for 14 % of the country area and have the highest population density and higher than average per-capita income levels. 'Partial deforestation' involve municipalities that have lost, on average, around 40 % of their forest area over the last century. They represent 15 % of the country area and about 4 % percent of the national forest area. Being mostly polarized along the elevation gradient, these municipalities show lower than average population densities and per-capita income levels close to the average. They also have higher than average population densities and, on average, the highest per-capita income values. Finally,

Table 2

Classification of Italian municipalities (n = 8101) and descriptive statistics using selected indicators.

Classification	A	B	C	D	E	Total
Number of municipalities	2176	1223	3127	1007	568	8101
Average municipal area (km ²)	1888	3290	5321	4394	1778	3729
Area (%)	13.6	13.3	55.1	14.7	3.3	100.0
Forest cover, 2018 (km ²)	0.0	5623.5	87,750.2	11,424.1	0.0	104,797.8
Forest cover, 2018 (%)	0.0	1.9	29.1	3.8	0.0	34.7
Share in forest cover, 2018 (%)	0.0	5.4	83.7	10.9	0.0	100.0
Change in forest area (%)	0.0	100.0	169.7	−42.9	−100.0	96.8
Average elevation (m)	171	339	480	437	312	358
North/Center (%)	76.8	49.3	66.0	71.5	85.4	68.4
South (%)	23.2	50.7	34.0	28.5	14.6	31.6
Coastal (%)	9.2	21.6	10.7	8.8	5.3	11.3
Population (%)	24.9	15.7	39.7	14.9	5.0	100.0
Pop. density (inhab/km ²)	258	172	120	133	204	173
Avg per-capita income (euros)	10,314	9344	9889	10,054	10,764	10,003

*Class (A), Municipalities without forests both in 1936 and 2018; Class (B), Municipalities with no forests in 1936 and with forest cover in 2018 (net increase), delineating a process of complete 'forestation'; Class (C), Municipalities with forests in both 1936 and 2018, and with absolute forest increment between 1936 and 2018, reflecting a 'partial forestation'; Class (D), municipalities with forests in both 1936 and 2018, and with forest decrease in 1936–2018, reflecting a 'partial deforestation'; Class (E), Municipalities with forest cover in 1936 and no forests in 2018, delineating a process of full deforestation.

municipalities that have undergone a process of 'complete deforestation' between 1936 and 2018 show the highest population density and constitute just 3 % of the country area.

3.1. Pair-wise correlation analysis

The maps of forest cover indicators (G1–G8) and forest change (H1–H8) are reported in SM.Fig. 1 and SM.Fig. 2. The indicators referring to forest cover (2018) show significant correlations with the socioeconomic status of municipalities, with different signs and intensity. The largest number of significant correlations (Fig. 2) involves the percent share of degraded forests (2018) in the landscape (41 background indicators), total forest cover (36) and beech cover (35). The highest positive correlations with forest cover are found for the following research dimensions: environment (43 % of total indicators), population and territory (22 %), agriculture (19 %), economy (14 %), labor market and human capital (2 %). As regards the environmental dimension, recurrent correlations with forest variables characterize the 'ecological quality' theme: E4 (topsoil organic carbon content), E6 (environmental pressure from grazing index), and E7 (landslide risk index). Other socioeconomic variables positively correlated with forest cover indicators include SR-P4 (the share of pastures and meadows in total agricultural land), SR-P6 (farm surface area covered by woodland), and the agricultural landscape dimension. The socioeconomic indicators negatively correlated with forest cover belong to the following themes: agriculture (37 %), population and territory (22 %), economy (15 %), environment (13 %), labor market and human capital (10 %), and quality of life (2 %). From this perspective, stable correlations were found with SR-A5 (share of agricultural utilized area in total agricultural land), SR-P8 (agricultural landscape diversity), and SR-M4 (crop intensity index). The highest correlation coefficients (positive or negative) were observed with agricultural, territorial, environmental, and demographic indicators. Oak-forest cover showed high correlations with indicators related with education, the job market, and population structure.

Considering pair-wise correlations between socioeconomic and forest cover change indicators, fewer significant coefficients were observed, and only for total forest cover (6 indicators with significant coefficients), conifers (5), and degraded forests (25). In the case of positive relationships, the most significant indicators concern environment (45 %), agriculture (22 %), population and territory (22 %), and economic (11 %) dimensions. The dimensions most correlated with forest change (positive sign) concern ecological quality, particularly indicators E6 (environmental pressure from grazing index), E3 (soil erosion risk index), and E7 (landslide risk index). Negative correlations were observed with indicators from the same themes: population and

territory (33 %), agriculture (28 %), economy (22 %), and environment (17 %). Significant correlations with D1 (annual population growth rate, 1921–1931), B5 (land quality index), S2 (density of workers per municipality surface area), and E2 (potential soil contamination footprint) have been also recorded.

3.2. Parametric and non-parametric canonical correlation analysis

Both analyses' runs indicate how the highest correlation coefficients were observed with rankings' matrices. In this case, Root 1 identified a particularly high correlation (0.78) between left (forest) and right (context) datasets. This outlines municipalities with positive values for the majority of forest cover indicators, with the exception of G4 (share of chestnut in total forest cover, 2018), positive values of indicators relating with human settlements, a high perimeter-to-area ratio, larger-than-average municipal area, and selected predictors of economic structure, such as the percentage of workers in the public sector or in hotels/restaurants. This last indicator also suggests a role for tourism, whose significant predictors include T1 (the ratio of beds in hotels and camping to resident population). Other positive values described along this root were associated with agriculture. More specifically, a positive correlation with land tenure (percent share of agricultural land owned by the state) and human capital (percent share of pastures and meadows in total agricultural land, and of farm area covered by woodland in total farm area) was observed.

Indicators of water use/management (number and capacity of reservoirs per 100 inhabitants), ecological quality (share of protected land in total municipal area, soil erosion risk, percentage of topsoil carbon content, environmental pressure from grazing, and landslide risk index), and environmental management (cost of fire-damaged forest restoration, and number of fire events in total municipal area) were also positively correlated with Root 1. Similarly, Root 1 was negatively associated with topographical and land dimensions (average municipal area and land quality index). A negative coefficient was also observed for annual population growth rate (1921–1931). Within the economic dimension, mention should be made of S2 and S6 (respectively, the density of workers per municipal area and the percentage of workers in manufacturing), and of T7 (ratio of resident population to the number of stores). Land tenure (SR-A5, agricultural utilized area in total agricultural land), agricultural landscape (SR-P2: arable land in total agricultural utilized area; SR-P8: agricultural landscape diversity), and crop intensity (SR-M1: number of agricultural machines per farm; SR-M2: irrigated land in total agricultural utilized area; SR-M4: crop intensity index) were also significantly associated with Root 1, together with E2, the potential soil contamination footprint. In general, Root 1 assigned

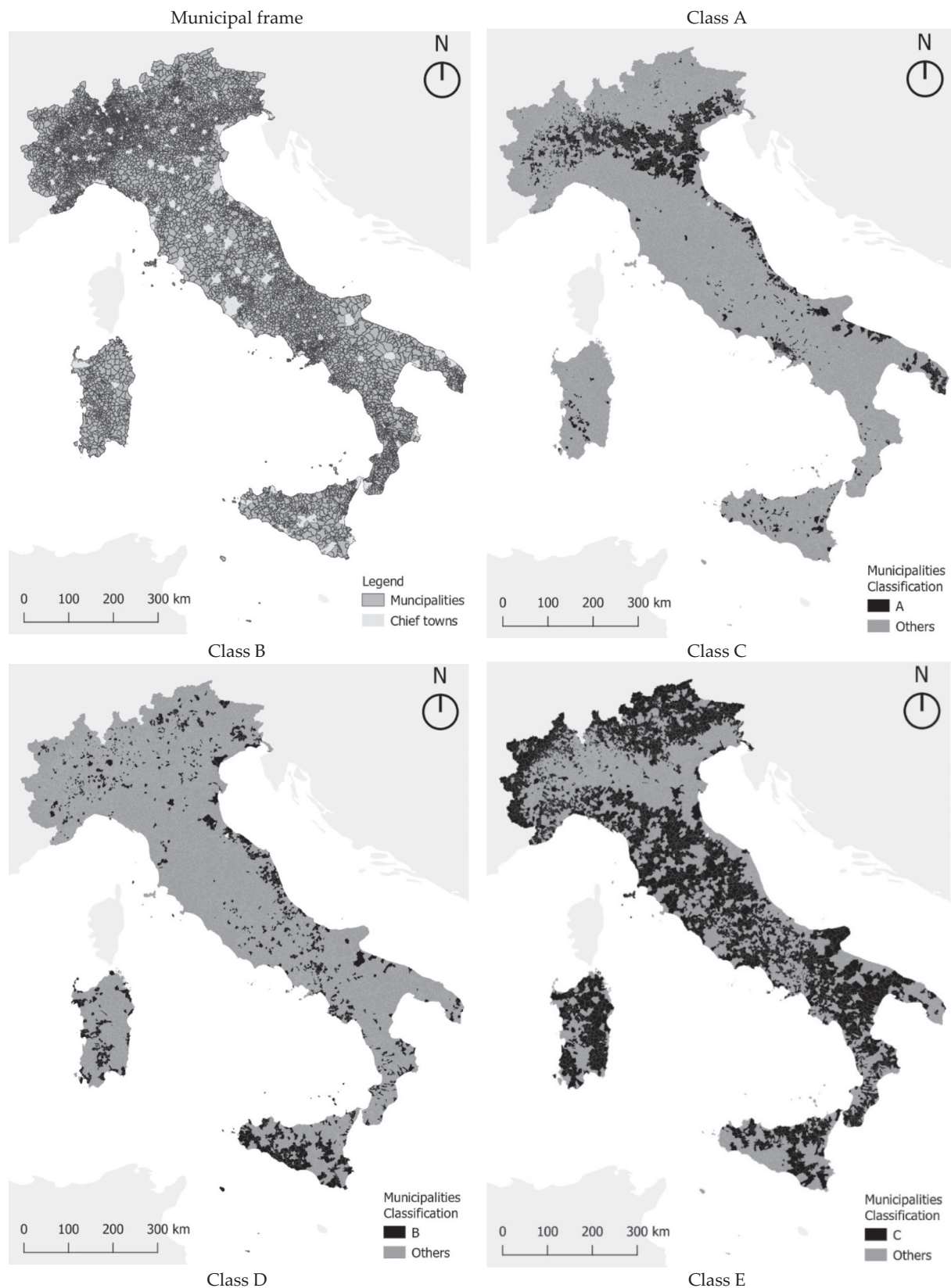


Fig. 1. Study area and the spatial distribution of each forest class.

positive coefficients to mountainous and hilly municipalities located in major mountain ranges, and negative coefficients to municipalities in flat districts (Fig. 3).

Forest cover change indicators associated to Root 1 show a less intense correlation with socioeconomic indicators (0.66 using rankings) than forest cover indicators; the analysis clearly outlined municipalities

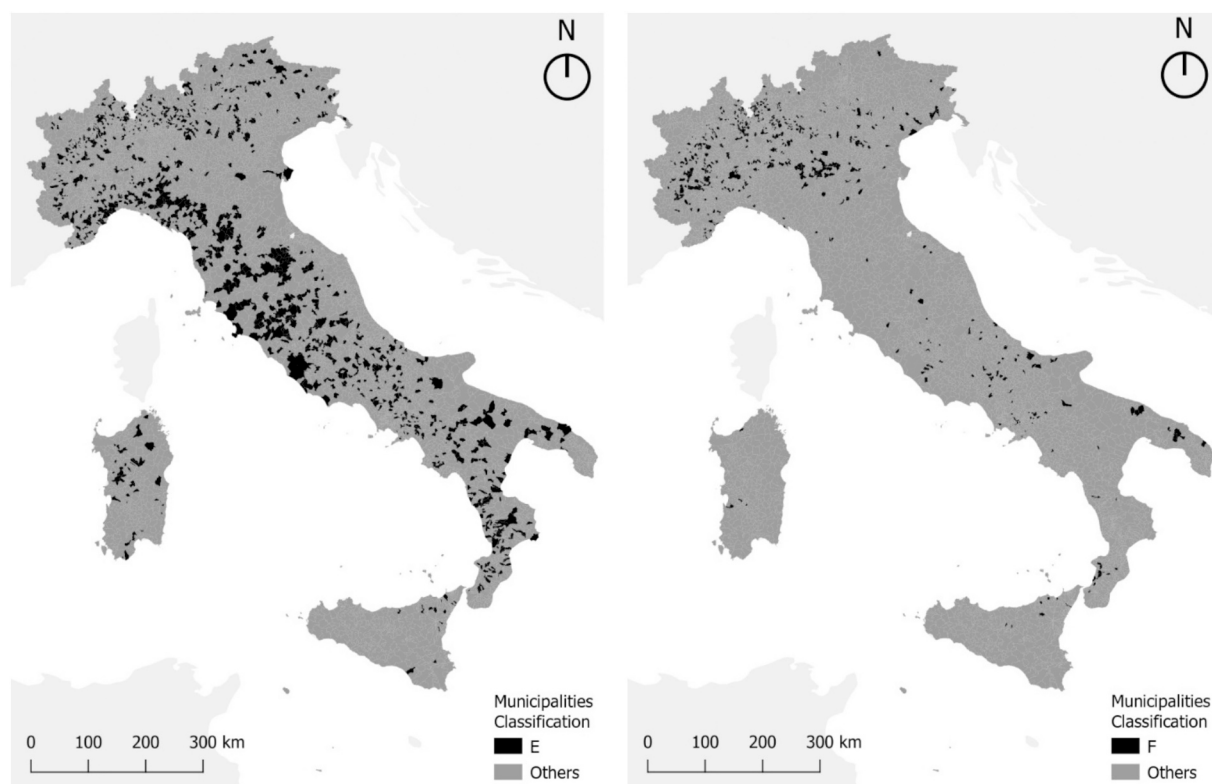


Fig. 1. (continued).

Table 3

Pair-wise correlations (based on Spearman coefficients) of forest cover indicators (G1-G8) or forest change indicators (H1-H8) and socioeconomic indicators (see Table 1 and SM. Table 1) in Italian municipalities ($n = 8101$); Percentage of correlative relationships ($p < 0.05$ after Bonferroni's correction for multiple comparisons) - divided by positive and negative - for each macro-area of socio-economic variables.

Indicators	Forest cover indicators (G1-G8)		Forest change indicators (H1-H8)	
	Positive Correlation index (%)	Negative Correlation index (%)	Positive Correlation index (%)	Negative Correlation index (%)
Population and territory	21.5	22.1	22.2	33.3
Labour market and human capital	2.2	10.5	0	0
Economy	14.0	15.1	11.1	22.2
Quality of life	0	2.3	0	0
Agriculture	19.4	37.2	22.2	27.8
Environment	43.0	12.8	44.4	16.7

with a net increase of forest area in some specific types (beech and others). These municipalities are positively correlated with the same indicators as found in the analysis of forest cover (2018). An important difference is the insignificance of W5 (number of reservoirs per 100 inhabitants) and E1 (percentage of protected land in total municipal area) along Root 1. Background indicators with negative coefficients are also similar with those found in earlier analysis, but highlight additional correlations, including D10 (annual population growth rate between 2011 and 2020); notably, new dimensions with significant correlations with forest change emerged. These include the following indicators: L1 (participation rate), L2 (activity rate), and L6 (female activity rate) of the job market dimension (labor market and human capital), Q2

(number of cars/inhabitants), and Q8 (per-capita amount of tax income) in the well-being dimension, and more generally, indicators assessing quality of life.

Considering the forest cover indicators, Root 2 presents a moderately high (multivariate) correlation (0.68) with the background indicators. In this case, oak forest cover is positively correlated with some indicators of 'labor market and human capital', especially F5 and F6 (respectively, the percentage of literate population without formal degrees and illiterate population in total population). From a spatial point of view, Root 2 highlights municipalities concentrated in Northern-Central Italy, basically away from the sea coast and dominant in the Apennine district. The same Root, in the case of forest cover change indicators, describes municipalities where forest diversification (Pielou J index) is associated with above-average percentages of population with secondary education. In this case, municipalities with positive values take on a complementary spatial distribution compared with the previous case (forest cover indicators), since they are concentrated on the islands, particularly Sardinia, and the Alps.

Root 3 shows a moderate correlation coefficient (0.49) between forest and background indicators and identifies municipalities with a net decrease of both chestnut and beech forest areas in Central-Southern Italy. Latitude, youth (< 35 years old) and female unemployment rates, percentages of population without a school degree and completely illiterate population, as well as average soil pH, are positively correlated with Root 3. Per-capita income tax and disposable income, the percentage of organic carbon content in surface soils, and soil compaction risk, have negative coefficients along Root 3. In the case of forest cover change indicators, Root 3 identifies a moderate correlation between forest cover change and contextual indicators (0.32), delineating municipalities with decreasing area of degraded forests, declining forest diversification and expansion in beech-forests. Soil erosion risk was found negatively associated with Root 3. The municipalities receiving positive coefficients to Root 3 are mainly located in central Italy and Northern Apulia.

Table 4

Results of a Canonical Correlation Analysis (CCA) of forest cover indicators (2018) and the socioeconomic context in Italian municipalities (n = 8101).

Variable	Standardized variables			Rankings			Description
	R1	R2	R3	R1	R2	R3	
<i>Correlation</i>	<i>0.49</i>	<i>0.34</i>	<i>0.31</i>	<i>0.78</i>	<i>0.62</i>	<i>0.49</i>	
Forest cover							
G1	0.48			0.83			Forest covered area (%)
G2	0.55	−0.52	0.60	0.87			Degraded forests (%)
G3				0.46			Other forest types (%)
G4						−0.65	Chestnut (%)
G5		−0.59	−0.65	0.75			Conifer (%)
G6				0.53		−0.50	Beech (%)
G7				0.46	0.83		Oak (%)
G8				0.84			Forest diversity index (J Pielou)
Socioeconomic and territorial context							
I2	0.54			0.58			Non-occupied houses (%)
I8	−0.49			−0.50			Average surface of dwellings (m2)
B1				0.63			Average municipal surface area (km2)
B2	0.50			0.59			Perimeter to Area ratio
B3			0.45			0.51	Latitude gradient (0: north-center, 1: south Italy)
B5	−0.71			−0.63			Land Quality Index
D1				−0.67			Annual population growth rate (%), 1921–1931
L2			−0.46				Activity rate
L3			0.41			0.41	Unemployment rate
L4			0.43			0.40	Unemployment rate of young people (< 35 years)
L6			−0.41				Female activity rate
L7			0.40			0.41	Female unemployment rate
L8			0.41				Unemployment rate of young women (< 35 years)
F5					0.46	0.45	Literate population without formal education degree (%)
F6					0.45	0.42	Illiterate population (%)
S2				−0.58			Density of workers per municipality surface area (km2)
S6				−0.40			Workers in manufacturing (%)
S10				0.45			Workers in hotels and restaurants (%)
S14				0.48			Workers in the public sector (%)
T1				0.46			Number of beds in hotels and campings / resident population
T7				−0.43			Resident population / total number of stores
Q8						−0.40	Per capita income tax amount (euros)
Q11						−0.40	Per capita disposable income (euros)
SR-A2	0.45			0.52			Agricultural land owned by the state (%)
SR-A5	−0.77			−0.64			Agricultural utilized area / Total agricultural land (%)
SR-L6	−0.40						Family farms (%)
SR-P2	−0.70			−0.53			Arable land / Agricultural utilized area (%)
SR-P4	0.69			0.58			Pastures and meadows / Total agricultural land (%)
SR-P6				0.68			Farm surface area covered by woodland (%on total farm area)
SR-P8	−0.75			−0.60			Agricultural landscape diversity (Shannon index)
SR-M1				−0.44			Number of agricultural machines per farm
SR-M2	−0.47			−0.46			Irrigated land / Total agricultural utilized area (%)
SR-M4	−0.69			−0.58			Crop intensity index
W5				0.41			Number of reservoirs (per 100 inhabitants)
W6				0.40			Reservoir capacity (per 100 inhabitants)
E1				0.41			Protected land (% in total municipal area)
E2				−0.62			Potential soil contamination footprint (equiv. inhab/km2)
E3	0.47			0.60			Soil erosion risk index
E4	0.71			0.55		−0.50	Topsoil organic carbon content (%)
E5						−0.50	Agricultural pressure to soil compaction index
E6				0.66			Environmental pressure from grazing index
E7	0.78			0.69			Landslide risk index
E8						0.49	Average soil ph
A1				0.42			Costs for (re)forestation of fire-affected woodland (euros per km2)
A2				0.43			Number of fire events in total municipal area

Insignificant predictors: I1,I3,I4,I5,I6,I7,I9,I10,B4,P1,P2,P3,P4,P5,P6,P7,P8,P9,P10,D2,D3,D4,D5,D6,D7,D8,D9,D10,L1,L5,L9,L10,L11, L12,L13,L14,F1,F2,F3,F4,S1, S3,S4,S5,S7,S8,S9,S11,S12,S13,S15,S16,T2,T3,T4,T5,T6,Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q9,Q10,Q12,Q13,C1,C2,C3,C4,C5,SR-A1,SR-A3,SR-A4,SR-A6,SR-Q1,SR-Q2,SR-Q3,SR-Q4,SR-Q5,SR-Q6,SR-Q7,SR-Q8,SR-Q9,SR-Q10,SR-L1,SR-L2,SR-L3,SR-L4,SR-L5,SR-P1,SR-P3,SR-P5,SR-P7,SR-M3,W1,W2,W3,W4,E9,E10,E11,E12,A3,A4,A5, A6.

4. Discussion

Based on the idea that long-term forest dynamics were asymmetrically distributed across Italy, our study examined forestation and deforestation patterns over a course of one century in relation to the socio-ecological and geo-economic structure of local areas (e.g. [Canadas and Novais, 2014](#)). Overall, forest expansion has resulted in a notable decline in forest fragmentation and patchiness, mainly through the

expansion of the existing wood patches ([Borrelli et al., 2014](#)). On average, forest patches have assumed a more geometrically regular shape, which has led to a substantial enlargement of forest core areas, as already highlighted by [Camarretta et al. \(2018\)](#). It has also been shown that connectivity has improved due to the expansion of habitat bridges between existing forest patches ([Kosmas et al., 2015](#); [Tomao et al., 2017](#); [Sallustio et al., 2018](#)). These trends have been consistently observed in several countries, both within and beyond Mediterranean environments

Table 5

Results of a Canonical Correlation Analysis (CCA) of forest change indicators (1936–2018) and the socioeconomic context in Italian municipalities (n = 8101).

Variable	Standardized variables			Rankings			Description
	R1	R2	R3	R1	R2	R3	
<i>Correlation</i>	<i>0.37</i>	<i>0.26</i>	<i>0.23</i>	<i>0.66</i>	<i>0.38</i>	<i>0.32</i>	
Forest change							
H1		−0.45	0.52	0.62			Forest covered area change (%)
H2			−0.76			−0.58	Degraded forest change (%)
H3	0.97			0.93			Other forest types change (%)
H4			0.40				Chestnut cover change (%)
H6			0.40	0.59		0.51	Beech cover change (%)
H8		−0.98			−0.86	−0.41	Forest diversity index (J Pielou) variation
Socioeconomic and territorial context							
I2	0.49			0.59			Non-occupied houses (%)
I8				−0.48			Average surface of dwellings (m2)
B1				0.60			Average municipal surface area (km2)
B2				0.56			Perimeter to Area ratio
B5	−0.49			−0.58			Land Quality Index
D1				−0.70			Annual population growth rate (%), 1921–1931
D10				−0.40			Annual population growth rate (%), 2011–2020
L1				−0.41			Participation rate
L2				−0.41			Activity rate
L6				−0.40			Female activity rate
F3					0.43		Population with secondary education (%)
S2				−0.61			Density of workers per municipality surface area (km2)
S6				−0.45			Workers in manufacturing (%)
S10				0.43			Workers in hotels and restaurants (%)
S14		−0.41		0.53			Workers in the public sector (%)
T1				0.40			Number of beds in hotels and campings / resident population
Q2				−0.40			Number of cars / inhabitants
Q8				−0.40			Per capita income tax amount (euros)
SR-A2				0.49			Agricultural land owned by the state (%)
SR-A5	−0.45			−0.54			Agricultural utilized area / Total agricultural land (%)
SR-P2	−0.49			−0.50			Arable land / Agricultural utilized area (%)
SR-P4	0.53			0.55			Pastures and meadows / Total agricultural land (%)
SR-P6				0.56			Farm surface area covered by woodland (%on total farm area)
SR-P8	−0.49			−0.54			Agricultural landscape diversity (Shannon index)
SR-M1				−0.53			Number of agricultural machines per farm
SR-M2				−0.42			Irrigated land / Total agricultural utilized area (%)
SR-M4	−0.53			−0.55			Crop intensity index
W6				0.40			Reservoir capacity (per 100 inhabitants)
E2				−0.62			Potential soil contamination footprint (equiv. inhab/km2)
E3	0.56			0.64		−0.44	Soil erosion risk index
E4				0.43			Topsoil organic carbon content (%)
E6				0.62			Environmental pressure from grazing index
E7	0.48			0.63			Landslide risk index
A1				0.41			Costs for (re)forestation of fire-affected woodland (euros per km2)
A2				0.42			Number of fire events in total municipal area

Insignificant predictors ($p > 0.05$): G5,G7,I1,I3,I4,I5,I6,I7,I9,I10,B3,B4,P1,P2,P3,P4,P5,P6,P7,P8,P9,P10,D2,D3,D4,D5,D6,D7, D8,D9,L3,L4,L5,L7,L8,L9,L10,L11,L12, L13,L14,F1,F2,F4,F5,F6,S1,S3,S4,S5,S7,S8,S9,S11,S12,S13,S15,S16,T2,T3,T4,T5,T6,T7,Q1,Q3,Q4,Q5, Q6,Q7,Q9,Q10,Q11,Q12,Q13,C1,C2,C3,C4,C5,SR-A1,SR-A3, SR-A4,SR-A6,SR-Q1,SR-Q2,SR-Q3,SR-Q4,SR-Q5,SR-Q6,SR-Q7,SR-Q8,SR-Q9,SR-Q10,SR-L1,SR-L2,SR-L3,SR-L4,SR-L5,SR-L6,SR-P1,SR-P3,SR-P5,SR-P7,SR-M3,W1, W2,W3,W4,W5,E1,E5,E8,E9,E10,E11,E12,A3,A4,A5,A6.

(e.g., [Galiana-Martin et al., 2011](#); [Raghavan and Shrimali, 2015](#); [Long et al., 2020](#)). The socio-ecological dimensions most correlated with forest change (positive sign) concern ecological quality (sensu [Quaranta et al., 2020](#)). However, geo-demographic forces (e.g. population growth rate, in particular in the time interval between the two World Wars, and the density of workers per municipality area) play a significant role, as proxies of human pressure and economic dynamics ([Garcia et al., 2020](#)).

The results of the descriptive analysis unequivocally demonstrate how long-term forestation (and deforestation) dynamics are distributed over space as a function of selected economic, socio-demographic and environmental characteristics. This is a relevant knowledge that expands the initial assumption that changes over time in forest cover are mostly associated with traditional geographical gradients, such as elevation, coastal-inland, and urban-rural axes (e.g. [Ferretti et al., 2014](#)). Spatially asymmetric dynamics in forest cover may reflect the assumption of Forest Transition Theory ([De Sy et al., 2012](#)), i.e. predicting that forest expansion follows different patterns and trends as a result of wealth accumulation, agglomeration/scale economies, and

socio-demographic dynamics ([Lambin and Meyfroidt, 2010](#)). We studied such dynamics with reference to (i) the total forest landscape and (ii) specific forest types, using multivariate exploratory techniques.

The present study identifies five long-term scenarios of forest dynamics based on the increase or decrease of forests ([Zambon et al., 2017](#)), taking municipalities as the appropriate analysis' domain. The empirical results show that forestation reflects a progressive land abandonment and the inherent decline in mountain economic systems, especially in the use of timber for building purposes (e.g. conifers and beech) and certain forest crops, such as chestnut ([Chianucci et al., 2016](#)). Over the Italian municipalities, the most frequent condition is the 'partial' expansion of forests: in addition to having per-capita income levels lower than the national average, these municipalities present peculiar characteristics of inland, peripheral communities with low population densities, especially in the mountainous districts of Northern Italy ([Ferretti et al., 2014](#)). In contrast, most of the municipalities that undergo a 'complete forestation' are close to the sea coast and located mainly in Central-Southern Italy, being associated with land

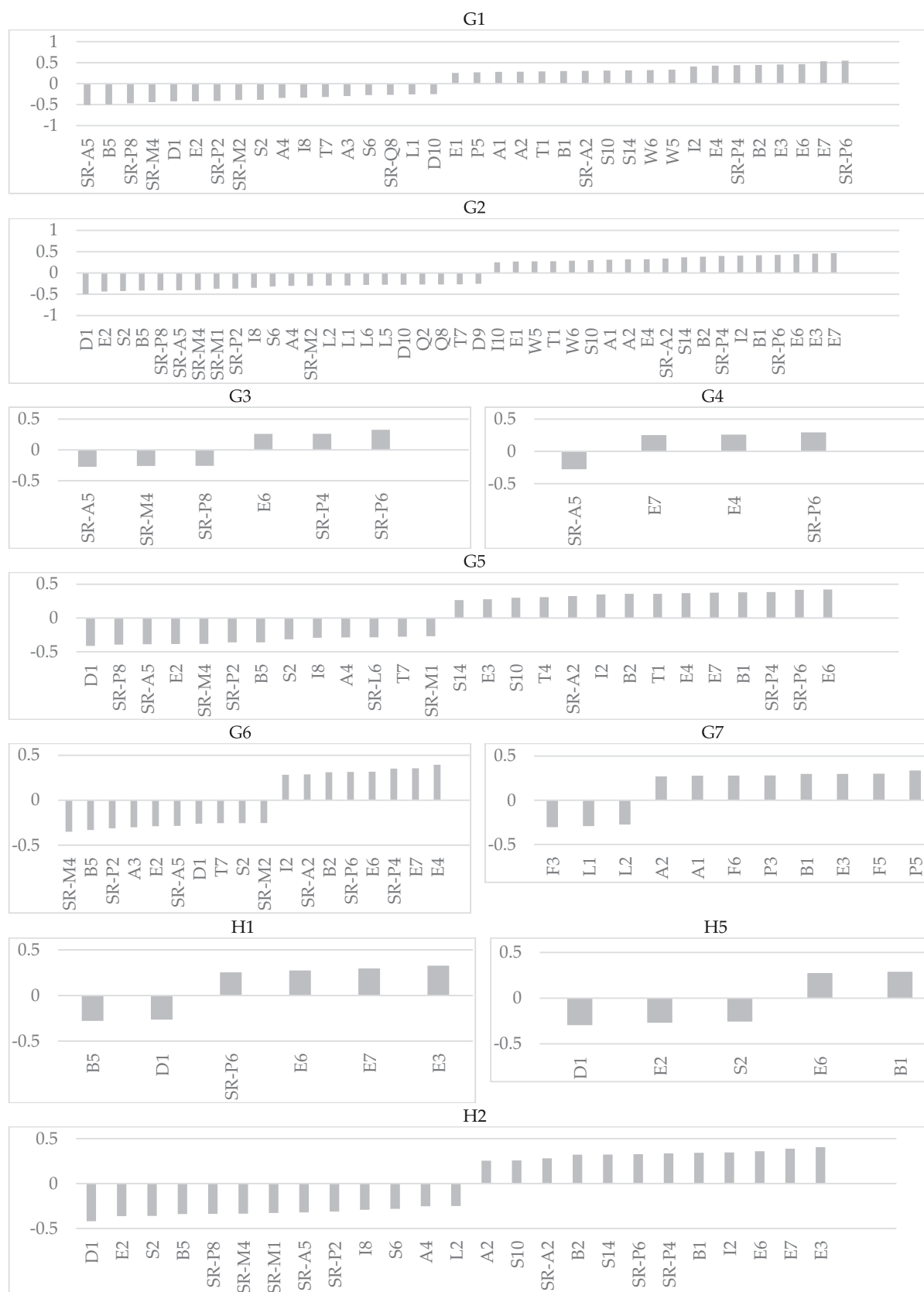


Fig. 2. Pair-wise correlations (based on Spearman coefficients) of forest cover indicators (G1-G8) or forest change indicators (H1-H8, Table 1 and SM.Table 1) and socioeconomic indicators in Italian municipalities (n = 8101); only correlations at $p < 0.05$ after Bonferroni's correction for multiple comparisons were shown.

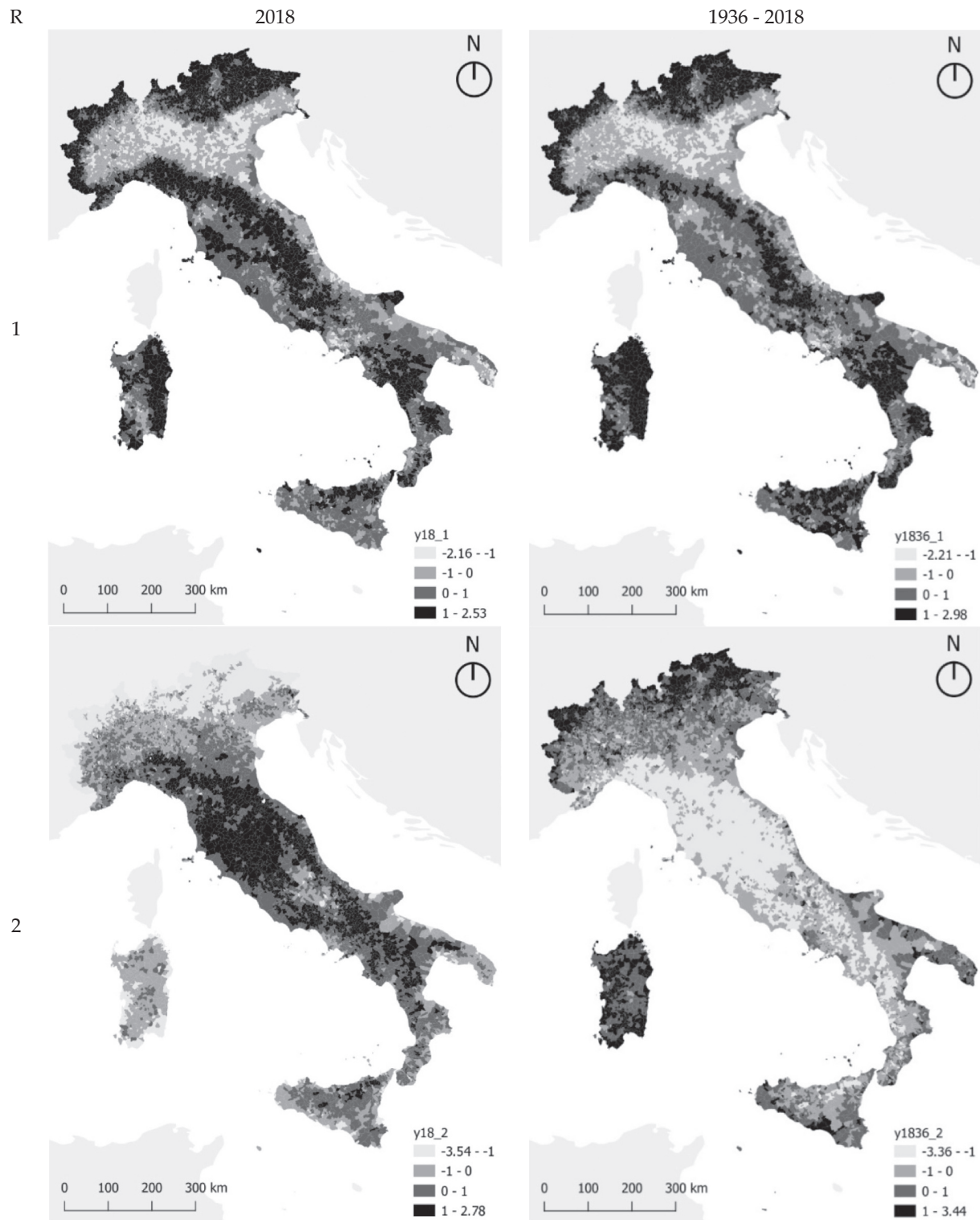


Fig. 3. Results of a Canonical Correlation Analysis (CCA) of forest cover and change indicators and the socioeconomic context in Italian municipalities ($n = 8101$). Spatial distribution of the canonical variates by root (R) and time lapse.

abandonment, dry farming and the economic unviability of alternative (non-forest) land use (Forino et al., 2015). Municipalities without forests in both 1936 and 2018 are mostly urban or, conversely, maintain their historical commitment to agriculture (Mancino et al., 2014, 2016). As regards the progressive loss of woodland, the case of 'partial deforestation' involves municipalities with no distinctive territorial

characteristics, which make identification of specific paths a difficult issue (Plieninger et al., 2011). On the contrary, the few municipalities undergoing a 'complete deforestation' present distinctive socioeconomic traits, being more frequent in Northern Italy and showing above-average economic characteristics and a greater population density (Smiraglia et al., 2015).

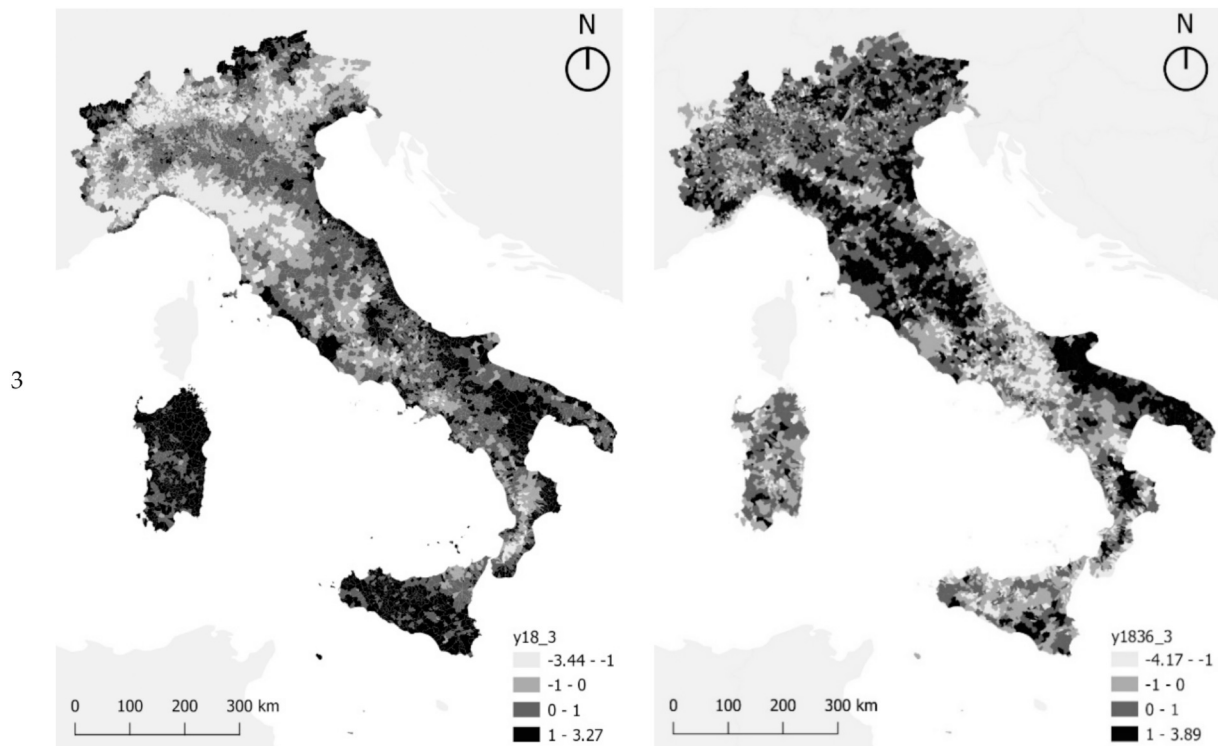


Fig. 3. (continued).

Deforestation was mainly dependent on urbanization and industrialization, especially during the 1950s and the 1960s (Salvati and Carlucci, 2016). Nowadays the conversion of forest land in Italy is legally regulated so that any change from forest to other uses should be usually compensated by placing a non-forest area of similar size under forest use (however, not necessarily in the same municipality). Nevertheless, legal deforestation processes due to settlements and infrastructure development may still be locally substantial (Corona et al., 2012). A simple correlation analysis showed the direct relationship between (the expansion of) forest cover and (the decrease in) agricultural land. Furthermore, forest cover increased in steep areas with low population density. Almost all forest indicators showed a negative correlation with the annual population growth rate between 1921 and 1931. This was indicative of the most intense phase of deforestation in Italy within the last century, being associated with a major push to demographic expansion and increased resource consumption (Scarascia-Mugnozza et al., 2000).

Canonical correlation analysis provided a summary vision of forest dynamics with respect to socioeconomic contexts, highlighting the dominance of non-linear relationships between the considered variables. This is consistent with the intrinsic characteristics of environmental-economic systems, which are increasingly complex, heterogeneous over space, and relatively unpredictable on both short- and long-time scales (Theobald, 2004). Furthermore, significant divides were observed between Northern and Southern regions, in accordance with the different levels of wealth in rural territories across Italy. In essence, long-term forest expansion in the last century appears as a recurrent phenomenon in different rural types (Raymond et al., 2009), probably representing – together with urbanization – the most frequent land-use transition in the whole Mediterranean basin.

By revisiting the prediction of the Forest Transition Theory in the case of advanced economies, our work clarifies how deforestation/forestation processes strictly depend on the background context, influencing the potential for sustainable development and the resilience of local territories (Plieninger et al., 2011). Being dominant above 500 m elevation, forest expansion has clearly been accompanied by a

progressive abandonment of agricultural land and a reduction in population density and human pressure (Mancini et al., 2018). Due to economic marginality and highly depressed social dynamics that prevent durable sustainable development of local districts (Otero et al., 2015), the expansion of forests – and the consequent reduction of ecological disturbance – is interpreted mainly as a consequence of economic backwardness, often incapable to lead to a balanced development path (Niedertscheider and Erb, 2014). From this perspective, not all forest expansion is considered positive.

As a general rule, in both Mediterranean and temperate environments, landscape changes are often perceived unfavorably, as they are associated with the progressive disappearance of traditional landscapes, loss of biodiversity, habitat quality, cultural heritage and place identity (e.g. Antrop, 2005). Although forest expansion is possibly regarded as a sign of improved environmental quality, its effects on biodiversity and landscape conservation may be either positive or negative (Cimini et al., 2013), depending on local ecological conditions and on the social perception of symbolic and cultural values (Mancino et al., 2016). For instance, the encroachment of mountain pastures is usually perceived negatively from a recreational perspective (Niemela et al., 2005). These considerations suggest that forest expansion requires careful land-use planning and management (Sallustio et al., 2018), especially in peri-urban areas (e.g. Colantoni et al., 2017) and in rural landscapes characterized by medium population density, high accessibility, and dynamic economic systems (Tomaio et al., 2017).

Our study, consequently, enhances and substantiates the rationale behind Forest Transition Theory, demonstrating innovation in both the analytical scale and the quantity of units employed, successfully capturing dynamics that are challenging to discern at the national level or through local case studies (MacDonald, 2023). This analytical approach facilitated the examination of the issue on a national scale while considering the heterogeneity of the Italian territory, where urban and rural situations necessitate distinct policy choices. Nonetheless, the work's potential is contingent upon the data utilized, as these are the primary datasets accessible at the investigated spatial scale (Salvati et al., 2017). The internal and external consistency of forest cover

indicators generated from various mapping sources or sampling designs has been examined in earlier studies. These studies have found that the choice of forest cover data sources has a key impact on the empirical findings, and that only few of them take long-term considerations into account because of the limited availability of frequent and comparable temporal observations. Even with regard to socioeconomic and spatial indicators, despite the effort in constructing such a large panel at such a detailed geographical scale, the availability of data from different time periods could even more clearly emphasize the relationships observed through the results of this study.

As final consideration, the results of this study confirm the practical importance of incorporating long-term forest landscape dynamics into land-use planning: spatially identifying such changes can help define priorities for forest management and enable more effective landscape governance (Corona and Alivernini, 2024). From this perspective, two exemplificative issues illustrate how past policies aimed at supporting Rural Development Programs (RDPs) – like those established at European Union (EU) level – have often fallen short, despite the fact that they remain major drivers of landscape change, at least in Europe. The first issue concerns peri-urban forests: these are recognized as a crucial priority in the EU policy agenda (European Environment Agency, 2006b), yet clear actions remain limited, and multi-criteria approaches are still required to foster cooperation between scientists, planners, and urban residents (e.g. Colantoni et al., 2015). Future planning strategies should therefore address challenges related to the practical containment of urban expansion and land fragmentation, for instance by promoting spatially coordinated networks that preserve the (relict) ecological functions of fringe landscapes (Salvati et al., 2017). The current mismatch between local-scale needs and policy frameworks highlights the need of integrating RDPs, urban development strategies, and environmental protection measures into a unique planning vision (Colantoni et al., 2017).

The second issue is framed under the relationship between increasing forest cover and wildfire prevention: Colonico et al. (2022) have highlighted a spatial mismatch between RDP expenditure on wildfire prevention and the areas most affected by fire activity, stressing the need for greater integration of territorial planning information into RDP funding criteria (Ferrara et al., 2017). Embedding indirect prevention measures within fire management plans would represent a cost-effective way to maximize the impact of public policies (e.g. Quaranta et al., 2020), allowing scarce resources to be directed towards high-risk areas, particularly those undergoing rapid forest expansion and increasing exposure to sprawling settlements (Ferrara et al., 2016). Along similar lines, the EU Green Deal may provide both the political support and the strategic framework (through mainstreamed EU strategies and funding opportunities) needed to replicate practical fire-smart models and functional approaches to wildfire risk prevention, as suggested by Ascoli et al. (2023).

5. Conclusions

Drawing on empirical evidence from Italy, this study supports the core assumptions of the Forest Transition Theory, a key research topic in forest assessment, policy and management, and highlights the role of socioeconomic forces, agricultural drivers and broader factors related to ecological quality shaping local dynamics of forest expansion or decline. The findings show that specific socioeconomic processes (and territorial profiles) reflect net gains (or loss) in forest area – with expansion largely understood as a (mainly natural) outcome of land abandonment, and decline intrinsically dependent on rising anthropogenic pressures. A distinctive contribution of this research lies in emphasizing territorial specificities of forest gain (or loss) dynamics, which are found to depend strictly on the economic and demographic characteristics of local communities, as well as on the degree of land protection. This aspect, together with the related considerations regarding forest functions, ecosystem stability, and land management (Führer, 2000), is relevant

not only for other Mediterranean (and European) countries but also for local areas beyond the Mediterranean region that share enough comparable conditions (e.g. Klooster, 2003; DeFries and Pandey, 2010; Long et al., 2020).

As an important line of investigation for future studies, a distinction should be made between passive and active dynamics, considering the outcome of targeted planning tools, e.g. for global change mitigation. From an environmental point of view, forest expansion should be evaluated on the basis of specific indications about active (or passive) forestation interventions. A more specific focus on planning constraints, e.g. quantifying the net impact of areas designated for nature conservation, infrastructure and/or settlement enlargement, is another research task that deserves further investigation. Fueling sustainable development in rural regions necessitates policies that integrate environmentally advantageous elements, such as forest expansion, to establish a more equitable socio-ecological framework for rural territory. This will help reduce depopulation and foster local economies through low-impact socio-demographic processes that are compatible with the environmental quality typical of 'forest dominated' landscapes. Forest expansion has occasionally been observed in peri-urban districts, impacting land management in broader metropolitan regions, first of all with respect to wildfire risks. Requiring completely different policy tools from those derived from a traditional 'rural development' perspective, these dynamics can benefit from a distinctive forest planning approach oriented towards urban greening and natural infrastructures.

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CRedit authorship contribution statement

Alessia D'Agata: Formal analysis, Data curation. **Marco Marchetti:** Project administration, Methodology, Investigation. **Luca Salvati:** Writing – original draft, Software, Resources. **Piermaria Corona:** Writing – review & editing.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.forpol.2025.103606>.

Data availability

Data will be made available on request.

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